Expense Tracker with Smart Overspending Alert

- **Theme- SDG 12 – Responsible Consumption**

  SDG 8 – Economic Growth

- **Category- Software/Hardware**
- **Group No - 8**
- **Group Members- • Soham Bhosale • Om Lad • Ojas Walke• Aryan Yadav**
- **Guide- Prof. Taruna Sharma**

Outline

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- Literature Survey
- Limitations of existing systems
- Problem statement and Objectives
- Proposed Methodology
- Experimental Setup
- Implementation
- Results & Analysis
- Conclusion and Future work
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Introduction

Background

- Students often fail to track daily expenses.
- Budget planning is usually ignored.
- Money shortage occurs before month-end.

Importance of Problem

- Financial awareness is essential for students.
- Helps reduce impulsive purchases.
- Promotes responsible spending habits.
- Supports SDG goal of responsible consumption.

Current Challenges & Motivation

- **Current Challenges**

• Manual expense tracking is inconvenient.
• Existing finance apps are complex.
• No prediction of future balance.
• Users do not know spending patterns.

- **Motivation**

• Students face end-of-month financial crisis.
• Need simple, student-friendly solution.
• Help users survive within monthly budget.

Objectives of Project

- Enable simple expense logging.
- Categorize expenses automatically.
- Calculate daily spending rate.
- Predict money exhaustion date.
- Provide smart overspending alerts.
- Track savings goals progress.
- Improve spending discipline.

Literature Survey

Observations

- Existing apps focus only on record keeping.
- No focus on student survival budgeting.
- Lack of intelligent alerts & predictions.

System	Feature	Limitation
Expense Manager Apps	Expense recording	No prediction
Banking Apps	Transaction logs	No budgeting
Wallet Apps	Reports	Complex UI

Limitations of existing Systems

- No future balance prediction.
- No intelligent spending alerts.
- No savings goal impact analysis.
- Complex financial dashboards.
- Manual data entry required.
- No student-focused financial planning.
- Limited behavioral spending insights.

Problem statement and Objectives

- **Problem Statement**
  Students lack a simple system to monitor expenses and predict future financial survival based on current spending patterns.

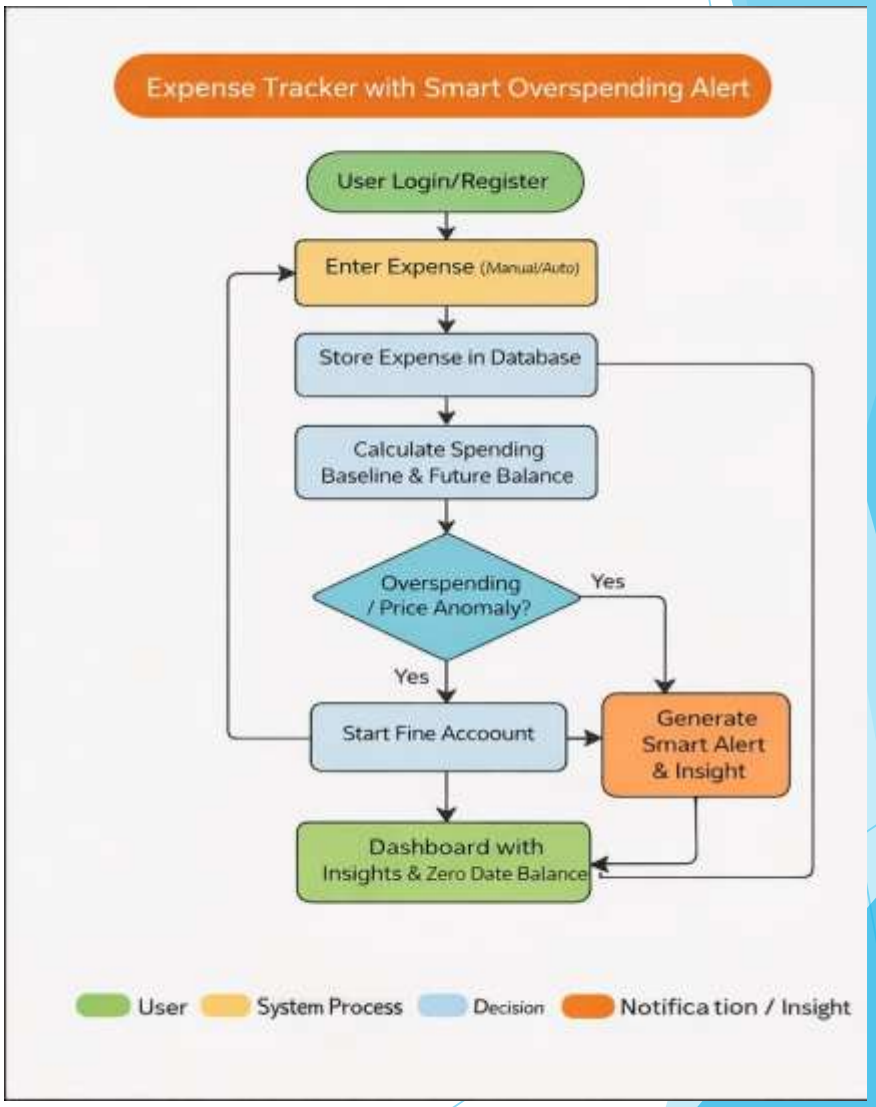
- **Objectives Addressed**

• Track expenses in real time.
• Analyze spending patterns.
• Predict survival days.
• Support savings goal planning.

Proposed Methodology

System Modules

- Expense Input Module
- Category Classification
- Firebase Database Storage
- Spending Analysis Engine
- Prediction Engine
- Dashboard Visualization
- Goal Tracking Module



Experimental Setup

Software Technologies

- React Js and React Native frontend
- Firebase Firestore database
- Firebase authentication
- Chart.js for visualization

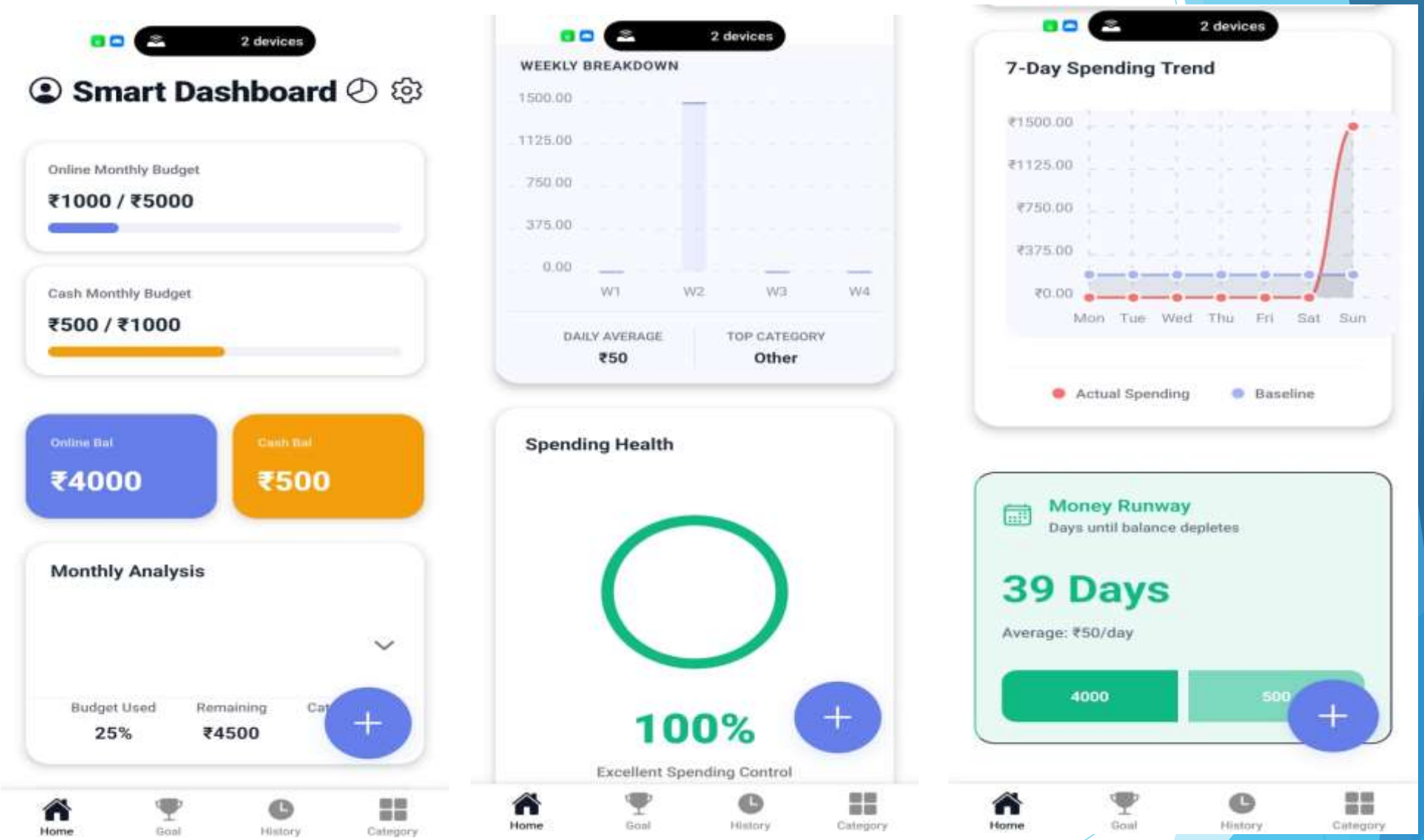
Tools Used

- VS Code editor
- Firebase console
- Browser testing tools

Input Data

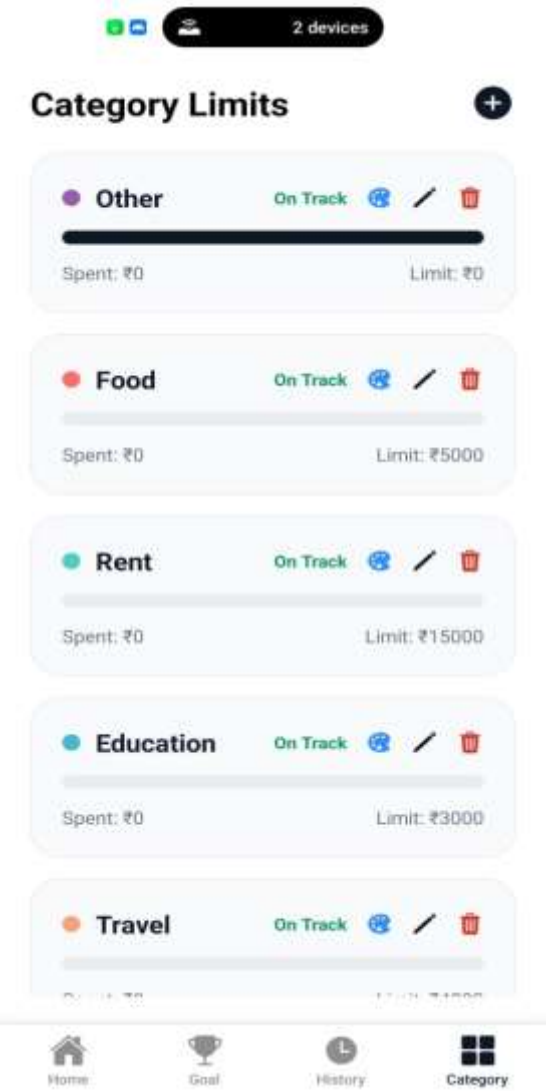
- User expense records
- Bank & cash balance inputs

Implementation

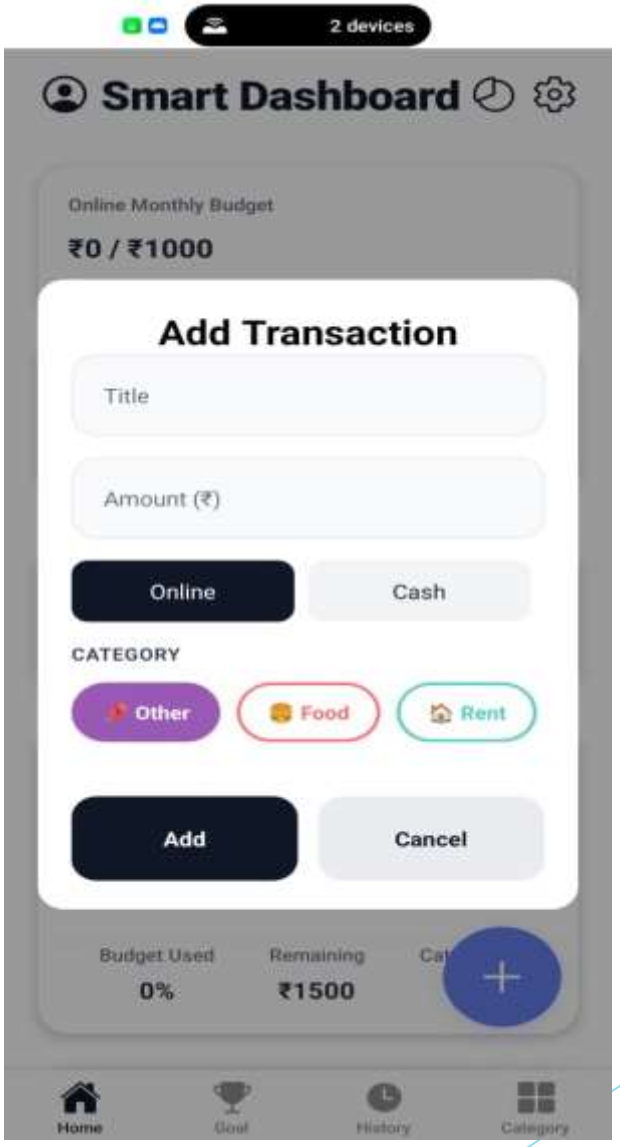


Dashboard

Implementation

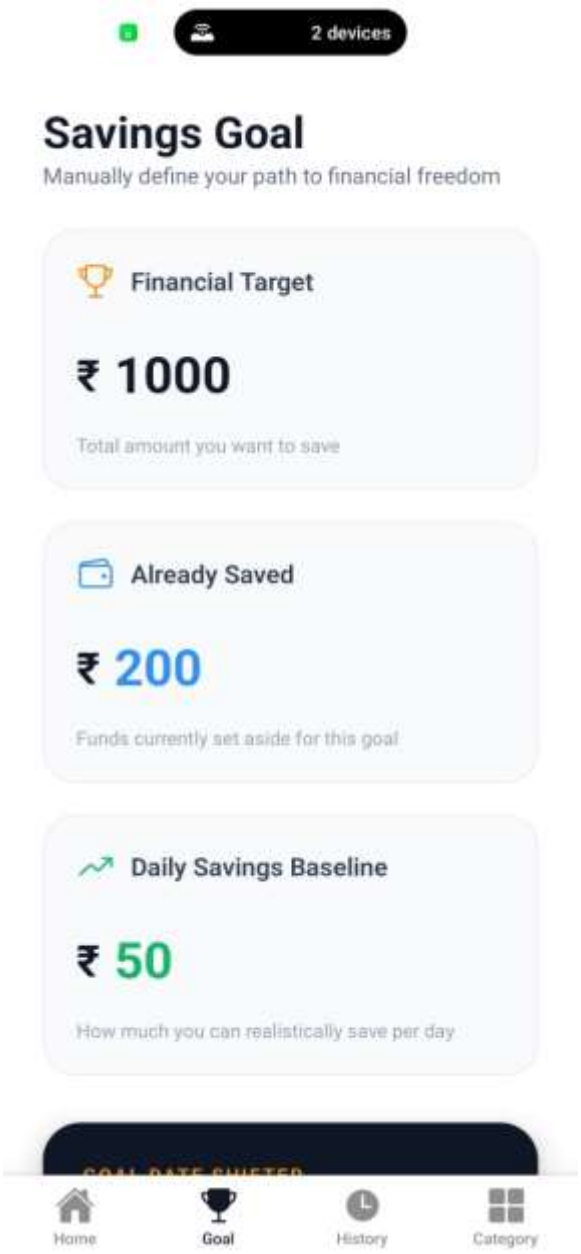


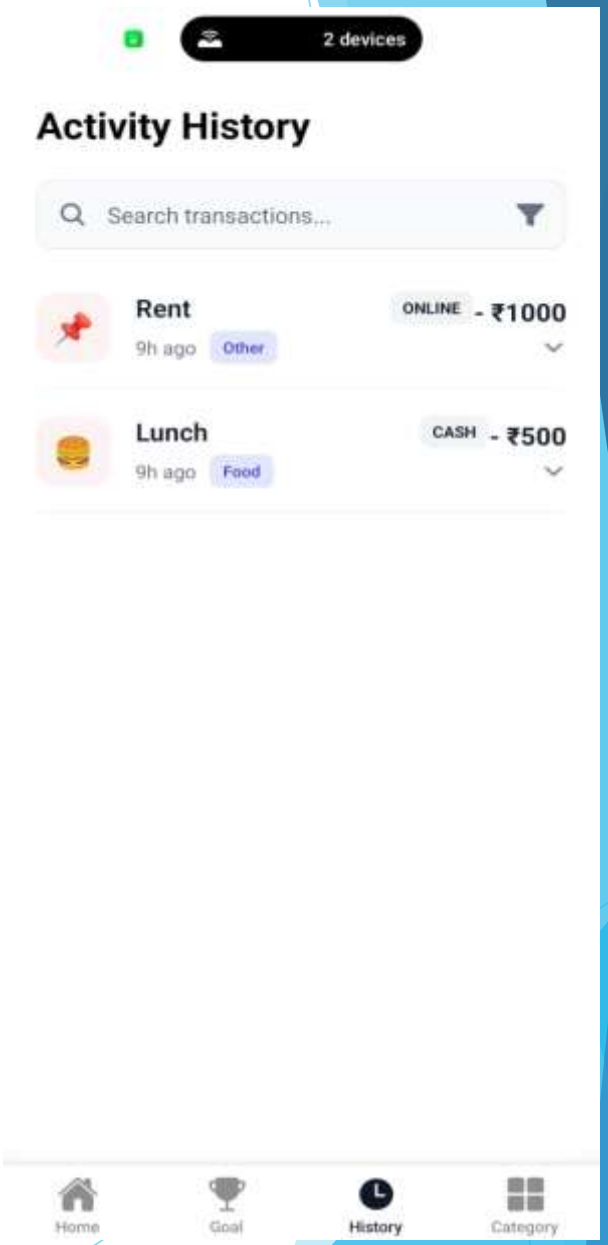
Categories



Add Transaction

Implementation





Results & Analysis

Observed Results

- Accurate daily burn rate calculated.
- Future balance prediction working.
- Spending trends visualized clearly.

Impact

- Users become aware of spending habits.
- Helps reduce impulsive expenses.
- Encourages savings behavior.

Conclusion

Outcomes

- Expense tracking simplified.
- Future survival prediction achieved.
- Smart alerts improve spending discipline.
- Student-friendly financial tool created.

Objectives Achieved

- Expense logging
- Spending analysis
- Budget prediction

Future Work

- **Improvements Planned**

• Automatic bank SMS transaction capture.
• Mobile app development.
• AI-based spending advice.
• Monthly budget recommendations.
• Notification alerts for overspending.
• Advanced goal prediction.

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Thank You...!!